

Task14 Report

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With this spike we had to play around with cohesion, separation, and alignment, using the previous Wander mode to simulate flocking behavior.

How it works:

Cohesion: with Seek mode, agents force their position towards their neighbors' average, pulling them together.

Separation: A force slightly pushes the Agents away from the others, to avoid visual overcrowding.

Alignment: A force matches the average heading of the neighbors, matching movement and direction.

It is written in the README file how to modify these values to experiment

****Cohesion:****

- `Z` : Decrease

- `X` : Increase

****Separation:****

- `C` : Decrease

- `V` : Increase

****Alignment:****

- `B` : Decrease

- `N` : Increase

****Wander:****

- M : Decrease
- σ : Increase

I managed to see multiple outcomes modifying the values of cohesion, separation, and alignment, for example, if we bump up the cohesion to 5.0 and bring all the other values to 0.0, you can see how all the agents collapse into a “ball” since there’s no separation and they overlap. Another example is bringing Alignment to 5.0, and everything else to 0.0, you can see how all the Agents move in the same direction, same speed, and they don’t group, they just move perfectly aligned.